U. 1. You are given some data (X_i, Y_i), $i = 1, 2, \ldots, n$ and asked to fit a straight line $\hat{Y} = b_0 + b_1X$ to it by least squares, and to perform a "full analysis." As you finish, the experimenter comes in and says "I've just found out all my X's are biased. Each X value I gave you is only 90% of the true X. Would you please redo the analysis." Using $1 - q$ instead of 90% (so you can handle the problem more generally) explain what parts of the "full analysis" are changed by this new information, and in what manner. (*Hint:* New $S_{XX} \equiv S^*_{XX} = S_{XX}/(1-q)^2$, etc., so $b_1^* = S^*_{XY}/S^*_{XX} = b_1(1-q)$, and so on.)

2. What would happen if the proportion differed for each observation, for example, was $(1 - q_i)$ for $i = 1, 2, \ldots, n$?

3. If in (1) or (2) the q or q_i were small, so that q^2 or q_i^2 could be ignored, how would the usual full analysis be changed?

V. (*Source:* "Life expectancy" by M. E. Wilson and L. E. Mather, Letter to the Editor, *Journal of the American Medical Association,* **229,** No. 11, 1974, 1421–1422.) The 50 pairs of observations below arose during a study carried out by Dr. L. E. Mather and Dr. M. E. Wilson. The variables are

X = Age of person at death (to nearest year);

Y = Length of lifeline on left hand in centimeters (to nearest 0.15 cm).

Many people believe that the length of one's life is linearly related to the length of one's lifeline. What light do these data throw on such a belief? You may assume that:

$$\Sigma X = 3333, \quad \Sigma X^2 = 231{,}933, \quad \Sigma XY = 30{,}549.75,$$

$$\Sigma Y = 459.9, \quad \Sigma Y^2 = 4308.57.$$

X = Age (yr)	Y = Length (cm)	X = Age (yr)	Y = Length (cm)
19	9.75	68	9.00
40	9.00	69	7.80
42	9.60	69	10.05
42	9.75	70	10.50
47	11.25	71	9.15
49	9.45	71	9.45
50	11.25	71	9.45
54	9.00	72	9.45
56	7.95	73	8.10
56	12.00	74	8.85
57	8.10	74	9.60
57	10.20	75	6.45
58	8.55	75	9.75
61	7.20	75	10.20
62	7.95	76	6.00
62	8.85	77	8.85
65	8.25	80	9.00
65	8.85	82	9.75
65	9.75	82	10.65
66	8.85	82	13.20
66	9.15	83	7.95
66	10.20	86	7.95
67	9.15	88	9.15
68	7.95	88	9.75
68	8.85	94	9.00

W. In *The Chicago Maroon* for Friday, November 10, 1972, The Party Mart advertised per bottle prices for vintage port as given in the accompanying table.

Article citat pel Draper i Smith (però les dades no hi són (?))

minished hearing, leukopenia, serum sickness, albuminuria, enuresis, urinary casts, fever, and neuritic pain.[2] Severe or unusual side effects have been precipitation of acute intermittent porphyria,[3] explosive exfoliative dermatitis, lupus erythematosus, photosensitive dermatitis, transient macular edema, and dysgeusia.

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1. Reaves LE III: Exfoliative dermatitis occuring in a patient treated with griseofulvin. *J Am Geriatri Soc* 12:889-892, 1964.
2. Steagall RW Jr: Severe reaction to griseofulvin. *Arch Dermatol* 88:218-219, 1963.
3. Berman A, Franklin RL: Precipitation of acute intermittent porphyria by griseofulvin therapy. *JAMA* 192:1005-1007, 1965.

Surgery Without Homologous Transfusion

To the Editor.—Patients who will not or cannot receive homologous blood transfusion for either elective or traumatic surgery present a dilemma. Attempts to control bleeding during surgical intervention are not new. However, performing major reconstructive hip surgery using blood-saving and retrieving techniques without the use of homologous blood transfusion is not a frequent procedure. Consequently, there is little or no information available.

In the past two years, we have successfully combined the use of hypotensive anesthesia and intraoperative autotransfusion in a group of patients whose religious beliefs prohibited homologous transfusion. Hypotensive anesthesia significantly lowered the amount of blood loss by 50% in this group of patients and aided in the insertion of acrylic cement. Also, 40% of the blood lost at surgery could be retrieved and replaced by intraoperative autotransfusion. Lesser amounts could be obtained in patients who bled little at surgery.

The data suggest that intraoperative autotransfusion provides little for the patient undergoing reconstructive surgery of the hip for the first time; when hypotensive anesthesia significantly reduced the surgical bleeding, the amount of blood retrieval was minimal.

These patients, barring any unforeseen circumstances, postoperatively sustain little drop in the hemoglobin level. On the other hand, the patient who has had *previous* reconstructive surgery of the hip poses a significant problem, as the blood loss at the time of surgery may be great. If one proposes to undertake surgery in a patient without the use of homologous transfusion, it appears mandatory that an autotransfusion apparatus and hypotensive anesthesia or some other alternative be available. If significant bleeding is encountered during surgery, intraoperative autotransfusion may be life-saving. Even in the patient who has not previously undergone this operation, an autotransfusion apparatus should be ready and primed for unexpected bleeding that may occur.

Although intraoperative autotransfusion and hypotensive anesthesia have been used successfully without major complications, it is difficult to draw general conclusions. Hypotensive anesthesia is an established procedure, but requires careful decision by an anesthesiologist and a surgeon for its use. Intraoperative autotransfusion for general use is still under investigation and is presently used for unusual problems.

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Acupuncture

To the Editor.—Given the public clamor for acupuncture therapy that John Bonica mentions in his overview (228:1544, 1974), the need for rigorous, clinical studies is clear. Frequently, acupuncture is the treatment modality of last resort, as is shown in the reports of six case studies of complications of acupuncture in patients whose chronic states had not responded adequately to conventional therapies and who, therefore, "resorted" to acupuncture (228:1552, 1974). While case studies of abuses are, of course, worth noting, they do not provide the invaluable evidence generated by more broad statistical analyses of larger samples, under carefully controlled conditions, as your EDITORIAL (228:1577, 1974) points out, that are required now.

In his review of acupuncture therapy, Bonica makes a historical error that, in fact, illuminates some of the current public and professional interest in acupuncture. Bonica says that the Kuomintang banned acupuncture in particular, and Chinese medicine in general, in 1929. As the English surgeon, J. S. Horn, reported in his account of 15 years in the People's Republic of China, *Away With All Pests* (New York: Monthly Review Press, 1969), p 71, an attempt was made by the Kuomintang to ban traditional medicine, and

> when this failed, they obstructed it by all kinds of restrictions and forced traditional doctors out of the cities. Nonetheless in their private lives many of the prominent Kuomintang leaders were devotees of traditional medicine. . . .

In his more detailed review of Chinese medicine, *The Chinese Art of Healing* (New York: Herder & Herder, Inc, 1971), Stephen Palos also refers to the attempt to quash acupuncture and traditional medicine (p 17), that Chiang Kai-shek initiated, only to retract under pressures made from the then capital of Nanking. As Marc Duke in *Acupuncture* (New York: Pyramid Publications, Inc, 1972), and Pierre Huard and Ming Wong in *Chinese Medicine* (New York: McGraw-Hill Book Co, 1968) relate, any attempt to restrict medical practice in China to Western-trained physicians would have effectively made medical care unavailable to most of the Chinese people, considering the scarcity of "modern" physicians in China. Given the needs of a population of half a billion people, some combination of Western, scientific, and traditional Chinese medicines was unavoidable. While there may be no analogous shortage of "modern" physicians in America to spur interest in acupuncture, the proliferation of acupuncture "mills" and clinics demands professional interest.

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Life Expectancy

To the Editor.—Life assurance companies need a reliable yet simple way to estimate life expectancy. Many palmists believe that the length of the life line can be used to predict life expectancy.[1]

We tested this idea in 51 cadavers of known age at death, using a map measurer to determine the length of the life line in each hand. The heel-crown length measurement enabled correction for differences in body size. Correlations between age and other variables (Table) were obtained through a matrix program (BMD 02D) using a CDC 6400 digital computer.

The critical value of correlation coefficient for significance ($P=.05$) is .27 ($N=51$). Within the Table, no value exceeded the critical value, and there-

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Lack of Correlation Between Length of Life Line and Age at Death*

	Age, yr	Length of Life Line, cm: Left	Left Height	Right	Right Height	Right and Left	Mean Height
Mean	65.2	9.2	0.056	9.4	0.058	9.3	0.057
SD	±16.7	±1.3	±0.004	±1.9	±0.012	±1.4	±0.009
Correlation coefficient with age	1.000	.056	.018	.021	.010	.040	.015

*This Table can be safely ignored by palmists.

fore no correlation was significant.

The life line was interrupted in one hand in six subjects; in these, the total length of that line was used in the calculations. Fortunately, the level of significance of the correlations was so low that it would have made no difference whether the six broken lines were included or not. These subjects died at the ages of 19, 40, 50, 75 and 82 years. Thus, a broken life line is not related to age at death, and it is our personal expectation that it correlates with nothing whatsoever.

This study is of the greatest importance because it is one of the few instances in which soothsaying has been, in sooth, objectively tested. We happily conclude that palmistry may be used to predict life expectancy but, when it is so used, it is blessedly free of scientific worthiness or usefulness to life insurers.

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1. Jones FW: *The Principles of Anatomy as Seen in the Hand*, ed 2. Baltimore, Williams & Wilkins Co, 1942.

Colonial Morphology Of *Neisseria gonorrhoeae*

To the Editor.—We have read recently with great interest the article by Brown and Kraus (228:862, 1974) regarding the recognition of the five colony types of *Neisseria gonorrhoeae*. The color photography truly enhances the morphological differences of the colony types.

Our observations agree with those of Brown and Kraus, and we are happy that, for the first time, there has been a clear indication in the literature of dark and light types 1 and 2. We have previously found type 1 and 2 gonococci to have variants, and we have designated the lighter variants types 1a and 2a. Under our experimental conditions, each of the colony types 1, 1a, 2, and 2a has reproduced true to the characteristics of the original types, and was shown, therefore, to be a *stable variant*.

Of interest to us, and not previously mentioned, is the fact that on prolonged incubation, types 1 and 2 (darker types) have produced only type 3, while types 1a and 2a (lighter variant) have generated only type 4. Furthermore, this pattern of intrastrain variation has been shown to occur with all of our type 1 and 2 colony variants.

Lastly, we have not been able to show morphological differences between types 1, 1a, 2, and 2a gonococci by electron microscopy. The pathogenicity of the aforementioned types is also essentially similar when assayed by (1) the intravenous infection of the chick embryo, and (2) the bactericidal effect of human serum.

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Heparin in Endocarditis

To the Editor.—Dr. T. W. Williams, Jr., in responding to a question (228:1297, 1974), stated that "there is now fairly general agreement that the use of anticoagulants is hazardous in patients with infective endocarditis." My colleagues and I agree that there is such agreement, but submit that the evidence on which such agreement is based is limited. Infected emboli are associated with hemorrhage, but available evidence does not show that anticoagulants make a difference in either incidence or severity of such hemorrhage.

Thus, it is our view that the common wisdom in this area needs reexamination. This is particularly so in patients with infective endocarditis who have a coincident indication for anticoagulant therapy (eg, venous thrombosis) and in patients with septic pelvic thrombophlebitis in whom heparin appears to have a positive therapeutic effect—yet in whom infective endocarditis also may be present.

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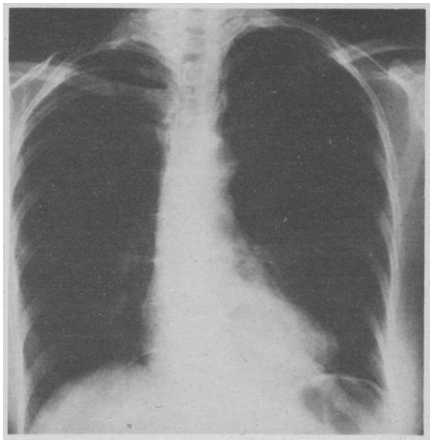
Pneumothorax as Complication of Acupuncture

To the Editor.—It is likely that pneumothorax is the most common, or the most easily recognized, potentially serious complication of acupuncture—at least as practiced in the western world. The recent article in The Journal (228:1552, 1974) by Carron et al describes two such occurrences, and cites as a reference a third (*Med J Aust* 1:941, 1973). The letters to the editor section of the *New England Journal of Medicine* (290:167, 1974; 290:633, 1974) provides two additional cases. A sixth patient, receiving her third treatment for atypical migraine in August 1973, noted left medial scapular pain and progressive dyspnea one hour following paravertebral insertion of acupuncture needles. When she sought medical attention 48 hours later, a 35% left pneumothorax was documented (Figure)



and treated with insertion of a chest tube. Re-expansion of the lung allowed removal of the chest tube on the third day and discharge from the hospital after four days. She has elected more conventional therapy for subsequent episodes of migraine.

While the case reports cited fail to mention whether relief of presenting symptoms ("painful shoulder," "systemic herpes," "low back pain") follows such therapeutic events, hospital stays of up to 14 days have been required for adequate treatment of complicating pneumothoraces.

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